Rishabh Tripathi

Boston, MA — +1 617-217-8673 tripathirishabh91@gmail.com www.linkedin.com/in/rishabh-tripathi

Summary

Bioinformatics Engineer building scalable NGS pipelines and multi-omics analytics using Nextflow and AWS (EC2/S3/Batch). Experienced in RNA-seq, ChIP-seq, and spatial transcriptomics with expertise in statistical modeling and interpretable machine learning. Develops reproducible, containerized workflows and interactive dashboards to translate high-throughput genomics data into actionable insight. Collaborative, analytical, and adaptable in cross-functional research environments.

Technical Skills

Programming: Python (Pandas, NumPy, Scanpy, scikit-learn, TensorFlow), R (Bioconductor, DESeq2, edgeR, Shiny), Bash, Linux
NGS & Genomics: Bulk RNA-seq, ChIP-seq, Small RNA-seq, Spatial Transcriptomics (Visium), WGS Variant Calling, Differential Expression, Pathway Enrichment, UMI handling, IsomiR profiling
Pipelines & Infrastructure: Nextflow, Docker, GitHub Actions (CI/CD), HPC environments, Reproducible workflow design
Bioinformatics Tools: STAR, HISAT2, BWA, GATK, HTSeq, FastQC, MultiQC, VCF/BCF/CRAM
Cloud & Visualization: AWS (EC2, S3, Batch), R Shiny, Highcharts, Matplotlib, Seaborn
Machine Learning: Elastic Net, XGBoost, Random Forest, SVM, SHAP, Leakage-safe validation, Feature engineering

Education

Northeastern University **Jan 2024 – May 2026**
Master of Science in Bioinformatics, GPA: 3.6 *Boston, MA*
Relevant coursework:Statistics for Bioinformatics, Genomics in Bioinformatics, Collect, Store, Retrieve, Analyze, Data, Omics, Transcriptomics.

Loyola Academy Degree and PG College **Aug 2020 – May 2023**
Bachelor of Science in Biotechnology, Chemistry, and Genetics, GPA: 3.1 *Hyderabad, India*
Relevant coursework: Plant Biotechnology, Molecular Biology, Agriculture Biotechnology, Immunology, Bioanalytical Techniques, Cell Biology

Experience

Bioinformatics CO-OP **July 2025 – Dec 2025**
Camp4 Therapeutics Corporation *Boston, MA*

- Designed and maintained reproducible NGS bioinformatics workflows in R, Python, and Bash, building custom data processing, QC, and visualization scripts and **deploying R Shiny web apps** for exploratory and summary-level analysis.
- Processed bulk **RNA-seq and ChIP-seq** datasets with production-grade Nextflow NGS pipelines on **AWS EC2**, supporting **time-course transcriptomics**, functional genomics, and gene regulatory analyses through parameterized, scalable workflow orchestration.
- Built and enhanced internal analytics and visualization tools for **target discovery** and **drug screening**, including interactive dashboards, EC50 **dose–response analyses**, **sequence-alignment visualizations**, and filterable UI, collaborating cross-functionally with scientists, data engineers, and chemistry team.

RNA Sequencing Pipeline Developer **Jan 2024 – May 2024**
Northeastern University *Boston, MA*

- Designed and optimized **a containerized small RNA-seq pipeline** modeled after *miRge3.0*, integrating **UMI-aware read collapsing**, **isomiR profiling**, and **parallelized alignment** to enhance analytical accuracy and achieve a **30% reduction in computational runtime**.
- Automated **NGS data processing** (QC, adapter trimming, alignment) using **Python**, **Bash**, and **Docker**, with continuous integration via **GitHub Actions** to ensure **reproducibility**, **traceability**, and **pipeline robustness**.
- Implemented **cloud-native sequencing workflows** leveraging **AWS S3** and **AWS Batch** for scalable compute and built R Shiny + Highcharts applications for **interactive visualization of miRNA/isomiR distributions and QC summaries**.

Projects

Cell-Type–Resolved Spatial Transcriptomics Atlas (Mouse Brain, Visium HD) **Dec 2025**
Spatial transcriptomics • scRNA-seq integration • Deconvolution • Domain analysis GitHub

- Built an **end-to-end spatial transcriptomics atlas** by integrating **Visium HD (8 µm & 16 µm) spatial RNA-seq data** with **single-cell RNA-seq references**, performing **cell-type deconvolution and label transfer** to generate **per-bin cell-type abundance maps** across tissue sections.
- Developed a spatial–expression domain pipeline with graph-based smoothing and Leiden clustering to detect coherent spatial domains and domain-specific marker genes via non-parametric differential expression.
- Implemented QC and enrichment analytics (UMI depth, gene complexity, mitochondrial content, domain-level enrichment) and exported standardized AnnData/Parquet/CSV outputs, interactive dashboards, and per-sample QC HTML for reproducible atlas generation

Interpretable Drug Response Prediction (DepMap / PRISM) **Aug 2025**
Interpretable Machine Learning • Genomics • Drug Response Modeling GitHub

- Built a leakage-safe ML pipeline to predict drug response from bulk RNA-seq gene expression (DepMap/CCLE) and PRISM primary screen data using Elastic Net and XGBoost.
- Applied **SHAP-based interpretability** and **pathway aggregation (GO/Reactome/KEGG)** to convert model predictions into mechanistic biological Insights, identifying key cholesterol and lipid metabolism drivers.
- Demonstrated **class-consistent biological explanations across four statins**, identifying shared driver genes (**SQLE, HMGCS1, GGPS1**) and conserved cholesterol-related pathways.

Single-Cell RNA-Seq Analysis & Cell Type Classification Pipeline **March 2025**
Python, Scanpy, scikit-learn, TensorFlow, GSEApy GitHub

- Built **scRNA-seq pipeline processing 10x Genomics PBMC** data through **QC, normalization, clustering and UMAP/t-SNE** using **Python** and **Scanpy;** engineered **cell type classification using Random Forest, SVM, and TensorFlow with SMOTE balancing**; implemented **GO and KEGG** pathway **enrichment** with GSEApy generating cross-cluster heatmaps and marker gene identification; **developed modular CLI for reproducible preprocessing, differential expression, and ML training workflows**.
- Optimized pipeline runtime to process 4,000+ cells through complete analysis workflow in 8 minutes on standard hardware.